

```
In [81]: import pandas as pd
```

```
In [82]: pd.__version__
```

```
Out[82]: '2.2.3'
```

```
In [83]: emp = pd.read_excel(r"C:\Users\LIKHITH\OneDrive\Documents\Rawdata.xlsx")
```

```
In [84]: emp
```

```
Out[84]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [85]: id(emp)
```

```
Out[85]: 2519210484560
```

```
In [86]: emp.columns
```

```
Out[86]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [87]: emp.shape
```

```
Out[87]: (6, 6)
```

```
In [88]: emp.head()
```

```
Out[88]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

```
In [89]: emp.tail()
```

Out[89]:

	Name	Domain	Age	Location	Salary	Exp
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderabad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [90]: `emp.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain      6 non-null      object
2   Age         4 non-null      object
3   Location    4 non-null      object
4   Salary      6 non-null      object
5   Exp         5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [91]: `emp.isnull()`

Out[91]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [92]: `emp.isnull().sum()`

```
Out[92]: Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

Data Cleaning

```
In [93]: emp['Name']
```

```
Out[93]: 0    Mike
          1    Teddy^
          2    Uma#r
          3    Jane
          4    Uttam*
          5    Kim
          Name: Name, dtype: object
```

```
In [94]: emp['Name'] = emp['Name'].str.replace(r'\W', '', regex=True)
```

```
In [95]: emp['Name']
```

```
Out[95]: 0    Mike
          1    Teddy
          2    Umar
          3    Jane
          4    Uttam
          5    Kim
          Name: Name, dtype: object
```

```
In [96]: emp['Domain']
```

```
Out[96]: 0    Datascience#$
          1         Testing
          2    Dataanalyst^^#
          3         Ana^alytics
          4         Statistics
          5             NLP
          Name: Domain, dtype: object
```

```
In [97]: emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex=True)
```

```
In [98]: emp['Domain']
```

```
Out[98]: 0    Datascience
          1         Testing
          2    Dataanalyst
          3         Analytics
          4         Statistics
          5             NLP
          Name: Domain, dtype: object
```

```
In [99]: emp['Location'] = emp['Location'].str.replace(r'\W', '', regex=True)
```

```
In [100]: emp['Location']
```

```
Out[100... 0      Mumbai
          1      Bangalore
          2      NaN
          3      Hyderabad
          4      NaN
          5      Delhi
          Name: Location, dtype: object
```

```
In [101... emp['Age'] = emp['Age'].str.extract('(\d+)')
```

```
In [102... emp['Age']
```

```
Out[102... 0      34
          1      45
          2      NaN
          3      NaN
          4      67
          5      55
          Name: Age, dtype: object
```

```
In [103... emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex=True)
```

```
In [104... emp['Salary']
```

```
Out[104... 0      5000
          1     10000
          2     15000
          3     20000
          4     30000
          5     60000
          Name: Salary, dtype: object
```

```
In [105... emp['Exp'] = emp['Exp'].str.extract('(\d+)')
```

```
In [106... emp['Exp']
```

```
Out[106... 0      2
          1      3
          2      4
          3      NaN
          4      5
          5     10
          Name: Exp, dtype: object
```

```
In [107... emp
```

Out[107...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [108...

```
clean_data = emp.copy()
```

In [109...

```
clean_data
```

Out[109...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

Misssing value treatement

In [110...

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain      6 non-null      object
2   Age         4 non-null      object
3   Location    4 non-null      object
4   Salary      6 non-null      object
5   Exp         5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [111...

```
clean_data.isnull().sum()
```

```
Out[111...] Name      0
          Domain    0
          Age       2
          Location   2
          Salary     0
          Exp       1
          dtype: int64
```

```
In [112...] import numpy as np
```

```
In [113...] clean_data['Age']=clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'])))
```

```
In [114...] clean_data['Age']
```

```
Out[114...] 0      34
          1      45
          2    50.25
          3    50.25
          4      67
          5      55
          Name: Age, dtype: object
```

```
In [115...] clean_data['Exp']=clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'])))
```

```
In [116...] clean_data['Exp']
```

```
Out[116...] 0      2
          1      3
          2      4
          3    4.8
          4      5
          5     10
          Name: Exp, dtype: object
```

```
In [117...] clean_data
```

```
Out[117...]
   Name  Domain  Age  Location  Salary  Exp
0  Mike  Datascience  34  Mumbai   5000    2
1  Teddy   Testing   45  Bangalore  10000    3
2  Umar  Dataanalyst  50.25    NaN  15000    4
3  Jane   Analytics  50.25  Hyderbad  20000  4.8
4  Uttam  Statistics   67    NaN  30000    5
5  Kim     NLP       55    Delhi  60000   10
```

```
In [118...] clean_data['Location']=clean_data['Location'].fillna(clean_data['Location'].mode()[0])
```

```
In [119...] clean_data['Location']
```

```
Out[119... 0      Mumbai
1      Bangalore
2      Bangalore
3      Hyderabad
4      Bangalore
5      Delhi
Name: Location, dtype: object
```

```
In [133... clean_data
```

```
Out[133...
   Name  Domain  Age  Location  Salary  Exp
0  Mike  Datascience  34  Mumbai   5000   2
1  Teddy   Testing  45  Bangalore  10000   3
2  Umar  Dataanalyst  50.25  Bangalore  15000   4
3  Jane   Analytics  50.25  Hyderabad  20000  4.8
4  Uttam  Statistics  67  Bangalore  30000   5
5  Kim     NLP      55    Delhi  60000  10
```

```
In [134... clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain       6 non-null      object
2   Age         6 non-null      object
3   Location     6 non-null      object
4   Salary       6 non-null      object
5   Exp         6 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
In [121... clean_data.to_csv('clean_data.csv')
```

```
In [140... import os
os.getcwd()
```

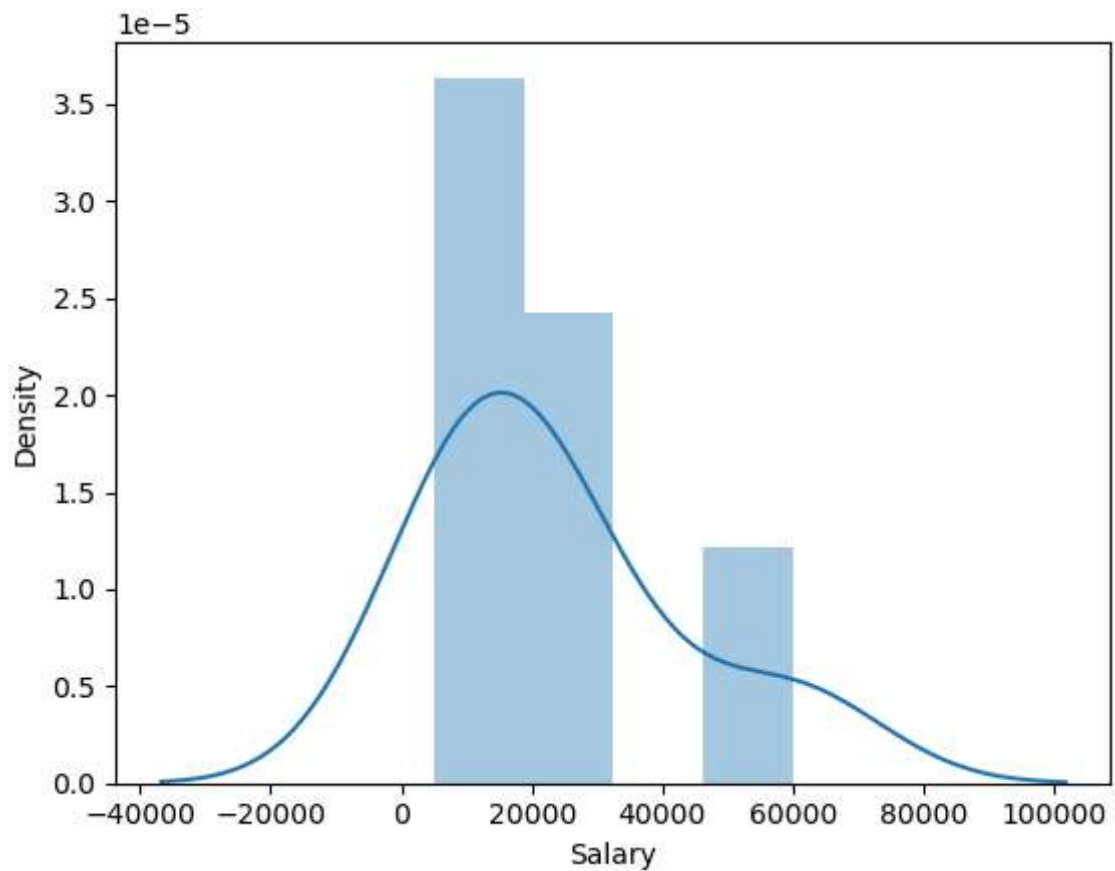
```
Out[140... 'C:\\Users\\LIKHITH\\Data Science'
```

EDA Technique apply

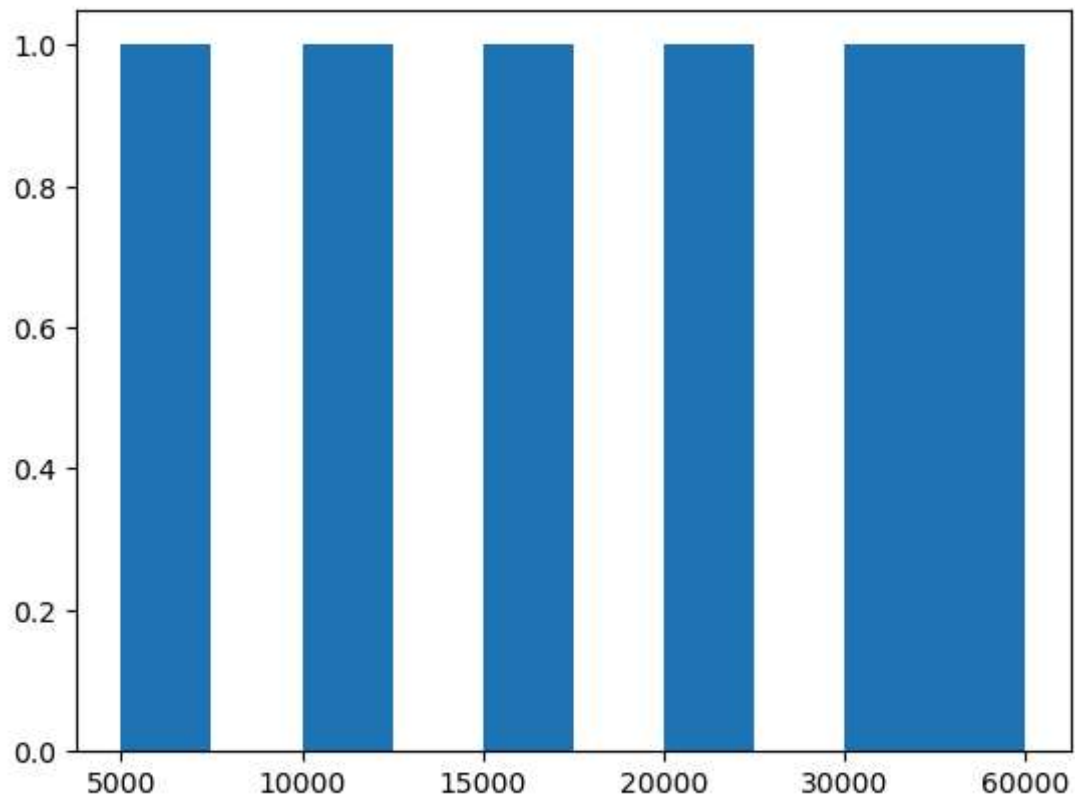
```
In [123... import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [124... import warnings  
warnings.filterwarnings('ignore')
```

```
In [125... vis1= sns.distplot(clean_data['Salary'])  
plt.show(vis1)
```



```
In [126... vis2=plt.hist(clean_data['Salary'])  
plt.show()
```

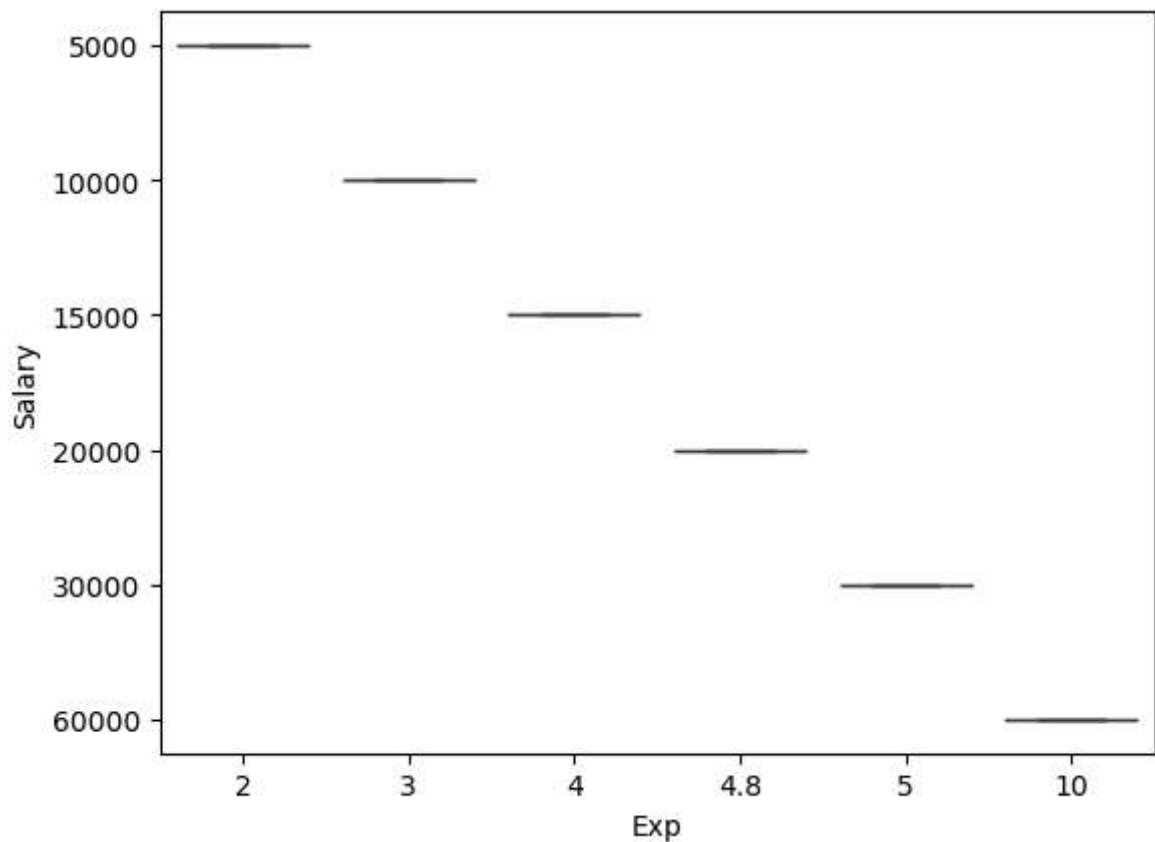



In [127...] `clean_data`

Out[127...]

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [128...] `vis3= sns.boxplot(data=clean_data, x ='Exp', y='Salary')`
`plt.show()`



In [129...] `clean_data`

Out[129...]

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [130...] `y=clean_data['Salary']`
`y`

Out[130...]

0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

Name: Salary, dtype: object

In [147...] `x_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]`
`x_iv`

Out[147...

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2
1	Teddy	Testing	45	Bangalore	3
2	Umar	Dataanalyst	50.25	Bangalore	4
3	Jane	Analytics	50.25	Hyderabad	4.8
4	Uttam	Statistics	67	Bangalore	5
5	Kim	NLP	55	Delhi	10

In [151...

```
y_dv = clean_data[['Salary']]  
y_dv
```

Out[151...

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [152...

```
y_dv
```

Out[152...

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [153...

```
emp
```

Out[153...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [156...

```
imputation=pd.get_dummies(clean_data, dtype=int)
```

In [157...

```
imputation
```

Out[157...

	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Name_Uttam	Domain_
0	0	0	1	0	0	0	
1	0	0	0	1	0	0	
2	0	0	0	0	1	0	
3	1	0	0	0	0	0	
4	0	0	0	0	0	1	
5	0	1	0	0	0	0	

6 rows × 33 columns



In [158...

```
clean_data
```

Out[158...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [141...

```
clean_data[:]
```

Out[141...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderabad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [142...

clean_data[0:6:2]

Out[142...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
4	Uttam	Statistics	67	Bangalore	30000	5

In [143...

clean_data[::-1]

Out[143...

	Name	Domain	Age	Location	Salary	Exp
5	Kim	NLP	55	Delhi	60000	10
4	Uttam	Statistics	67	Bangalore	30000	5
3	Jane	Analytics	50.25	Hyderabad	20000	4.8
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
1	Teddy	Testing	45	Bangalore	10000	3
0	Mike	Datascience	34	Mumbai	5000	2

In [144...

clean_data.columns

Out[144...

Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')

In []:

In []:

In [172...

In []:

In []: